

SMART FACTORY AUTOMATION SYSTEM DESIGNED IN CISCO PACKET TRACER

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ABSTRACT

This project addresses the design and simulation of an automated factory environment incorporating a broad spectrum of IoT devices, all governed through a centralized control panel system. It integrates critical industrial components—such as solar energy storage modules, RFID card readers, temperature regulators, automated conveyor belts, weighing sensors, motion sensors, and a fire protection network—into a single, coherent framework facilitated by Cisco Packet Tracer. By uniting networking protocols, embedded systems, and automated processes, the overarching objective is to illustrate how these converging technologies can boost operational efficiency, optimize resource utilization, and ensure top-tier safety in contemporary industrial settings.

In constructing this digitally interconnected infrastructure, the project places a strong emphasis on security protocols, real-time monitoring, and the seamless exchange of data among heterogeneous devices. Each stage of the factory’s workflow—from energy management and secure access control to automated material handling and emergency response—has been meticulously tested under simulated production scenarios to validate both reliability and interoperability. The proposed solution underscores clear advantages: reduced manual intervention, improved energy efficiency, and instantaneous adaptation to environmental factors. Concurrently, it highlights potential vulnerabilities, particularly the reliance on consistent network connectivity and the critical need for robust cybersecurity strategies to fend off malicious threats or unauthorized access. Taken in sum, the project demonstrates how an IoT-driven, automated factory can achieve enhanced productivity and resilience, establishing a practical roadmap for the future evolution of industrial automation while paving the way for further innovations and scalability.

Keywords

Automated factory, IoT, Cisco Packet Tracer, energy efficiency, RFID, conveyor belts, security

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ACRONYMS

IoT	–	Internet of Things
RFID	–	Radio Frequency Identification
WPA	–	Wi-Fi Protected Access
AES	–	Advanced Encryption Standard
SCADA	–	Supervisory Control and Data Acquisition
PLC	–	Programmable Logic Controller
VPN	–	Virtual Private Network
OSPF	–	Open Shortest Path First
RIP	–	Routing Information Protocol
BGP	–	Border Gateway Protocol
SAE	–	Simultaneous Authentication of Equals
EIGRP	–	Enhanced Interior Gateway Routing Protocol

1. INTRODUCTION

1.1. CONTEXTUALIZATION

The original idea for this project has developed from the increasing need to design and provide simulations for industrial environments where multiple IoT-enabled devices act cooperatively. Modern industrial environments are including more and more smart technology to perform tasks such as process monitoring and optimization, energy management, material handling, access control, and safety mechanisms. Though there are theoretical frameworks for such automated factories, a practice-oriented demonstration like the Cisco Packet Tracer would provide its own hands-on perspective for integration, testing, and refining ideas.

This work builds on knowledge already available in the field of industrial IoT (IIoT) and computer networking. It aims at taking practice further still by showing how the design and testing of robust intercommunication systems can be done in virtual laboratory environments while reducing the risks of implementation and highlighting defects before deployment. It is thus part of the overall vision of Industry 4.0, which sweeps in automation with connectivity and data-driven decision making in manufacturing and industrial settings.

1.2. NEED FOR IOT DEVICES IN INDUSTRY

Commercial impact through these IoT devices in industries has changed a lot in today's modern manufacturing and production systems. Organizations are embedding the following within machines and infrastructure:

- **Improvements in Operations Efficiency:** Automatic data collecting and real-time monitoring access can easily discover the bottlenecks and anomalies in operations, minimizing downtime and related operational costs.
- **Improving Decisions:** The trends of continuous sensor device data into analytics platforms would reinforce the depicted evidence and assist in schedules for predictive maintenance.
- **Improve Safety and Security:** An intelligent safety system that combines motion sensors and fire detection would proactively identify threats, while good networking protocols would assure secure data transfer.
- **Help with Sustainability:** Energy use information from IoT sensors could help optimize the use of resources in our facilities, reduce waste, and comply with environmental regulations.
- **Scalability:** Because of modularity, scalability is assured in the IoT architecture as new devices, lines of production or even technology are easily included as the nature of operations changes from time to time.

Integration of IoT devices into a connected network will transform isolated applications or manually monitored applications into smart self-regulated, self-adaptive, and continued improvement ecosystems. This project captures with an end-to-end demonstration how an organized IoT environment would have improved manufacturing performance, cost effectiveness, and sustainability, using Cisco Packet Tracer. [1]

1.3. OBJECTIVES OF THE PROJECT

The prime objective of this project is to create a working model for an automation factory, indicating how many numbers of IoT devices could be used in working as a single unit under a centralized management system. This objective is subdivided into smaller subgoals for effective processing.

- **Design and Configuration:** Model an integrated network with Cisco Packet Tracer involving devices for an automated manufacturing.
- **Security and Dependability:** Built security policies- ex. encrypted data transfer, access control list - and tested them to prevent unwarranted access to critical operational key data.
- **Real-Time Monitoring:** Create a central dashboard that shows real-time values on temperature, energy consumption, operation status of the conveyor, and safety alerts.
- **Validation and Analysis:** The : Measure system performance, reliability, and scalability under varying simulated conditions, identifying strengths and areas for improvement.

2. HISTORY AND EVOLUTION OF AUTOMATED FACTORIES

2.1. **EARLY DEVELOPMENTS: MECHANIZATION AND THE INDUSTRIAL REVOLUTION**

Industrial automation was rooted in the 18th-century Industrial Revolution, whereby technology replaced human labor in the act of manufacturing. The steam engine or more appropriately the highly improved steam engine of James Watt, turned factories upside down by driving heavy machinery harder, faster, and longer, and henceway increased massively the mass production with a reduction of dependence on man-made means of production as spinning and weaving among the rest. Such specialized machines, like the spinning jenny, would go a long way in simplifying repetitive processes since it heralds the emergence of modern factory automation.

Moreover, this meant far-reaching social change: workers migrate to urban centers, and to the thousands of standardized parts that made maintenance and repairs easier. The assembly line would become more prominent later, most notably at Henry Ford's automotive plants, but efficiency, consistency-of mass machinery- had already been planted in the Industrial Revolution, creating the roots for Industry 4.0's entirely integrated and information-driven manufacturing environments..

2.2. THE 20TH CENTURY: EMERGENCE OF INDUSTRIAL AUTOMATION

The whole thing was initiated from the beginning of the industrial automation in the 20th century. Developments that produced excellent propellers to enhance production in factories and which formed the foundation of the modern industrial practices. It was Henry Ford who introduced the dynamic assembly line in 1913 for automobile production and people began to realize the mass power- From the time reduced, span from saved the to scenes the of amount mass of production expense to the intensive use of these standard, repetitive tasks assigned to a specialized workstation.

But the age of electricity expanded possibilities for automation. All applications have simple control processes and reduced wiring through PLCs or programmable logic controllers that replace large relay panels. There was a shift from CNC machines that could produce parts with much greater accuracy and flexibility to slight automation, where "factories were allowed to hold higher-quality production with much less human intervention" in the automotive, aerospace and consumer electronics industries through a feedback loop and early digital systems. In the end, all these developments had a good basis for emerging into quite complicated and interdependent ecosystems of the last quarter of the twentieth century and the subsequent period.

2.3. FROM COMPUTER-INTEGRATED MANUFACTURING TO EARLY “SMART” SYSTEMS

Computer-Integrated Manufacturing has pioneered the second half of the 20th century literally. CIM, short for Computer-Integrated Manufacturing, unifies design, production, and quality control as a whole domain under digital framework. In other words, it allows people to share and feedback data between various departments on resource utilization and strict controlling product synopsis such as design specifications to final product inspections during various phases of manufacturing rather than in their separate independent processes. Hence, factories are able to operate on the computer networks managing the data coming in from design specifications to the finished products during the inspections.

Robots would then emerge as a major force on the production floor. The versatility of robots can be most impressive to human workers by doing work that is too dangerous, dull, or

precise for human handling. The early industrial robots begin to take over assembly production, welding, cutting, and placing parts as close as possible to the needed spaces without the touch of human hands. The incorporation of programmable robotics with computerized monitoring has reduced variability, opened up savings in labor cost, and increased general workflow efficiency for manufacturers by further bridging and erasing tedious manual processes.

One of the hallmark manifestations of this phase comprised Supervisory Control and Data Acquisition (SCADA) systems whereby real-time data could be obtained for visibility on the machine performance and production bottlenecks. SCADA system introduced a new dimension in terms of remote monitoring and control which suited the greater needs for high speed, large production lines. While these facilities were mostly limited to geographical areas-they were not too widely distributed, and were built for limited data analytical purposes-they laid a very firm foundation for the centralized process supervision, a concept that would evolve into modern IoT and cloud-based applications. CIM, robotics, and SCADA set the pace for what is being termed "smart" capabilities.

2.4. THE INDUSTRY 4.0 PARADIGM AND SMART FACTORIES

Industrial The Industry 4.0 Paradigm and Smart Factories Industrial automation enters into a completely new face of the 21st century with such convergence of CPS, IoT technologies, and advanced data analytics. This is commonly regarded as the fourth industrial revolution and is fundamentally a change of everything beyond that of replacement of human labor with machines. It envisages digitally interconnected ecosystems in which every component-machines, sensors, workers, and even products-communicate in real time. Such networks allow for intelligent decision-making at many levels-from on-the-fly creation of production schedules to predictive maintenance algorithms that envision machine breakage in advance.

Fundamentally at the Industry 4.0 model, there are smart factories providing cloud computing and processing at the edge to accommodate millions of operation data and re-inject them into the manufacturing environment. These factories will use autonomous robots,

digital twins, and machine learning to achieve dynamic workflow adaptations, downtime minimization, and personalization of output on the mass customization line. Industry 4.0 provides a comprehensive, data-driven framework fostering constant innovation, optimization of resources, and improved competitiveness for the contemporary manufacturing environment by integrating formerly isolated processes-from supply chain to customer feedback. [2]

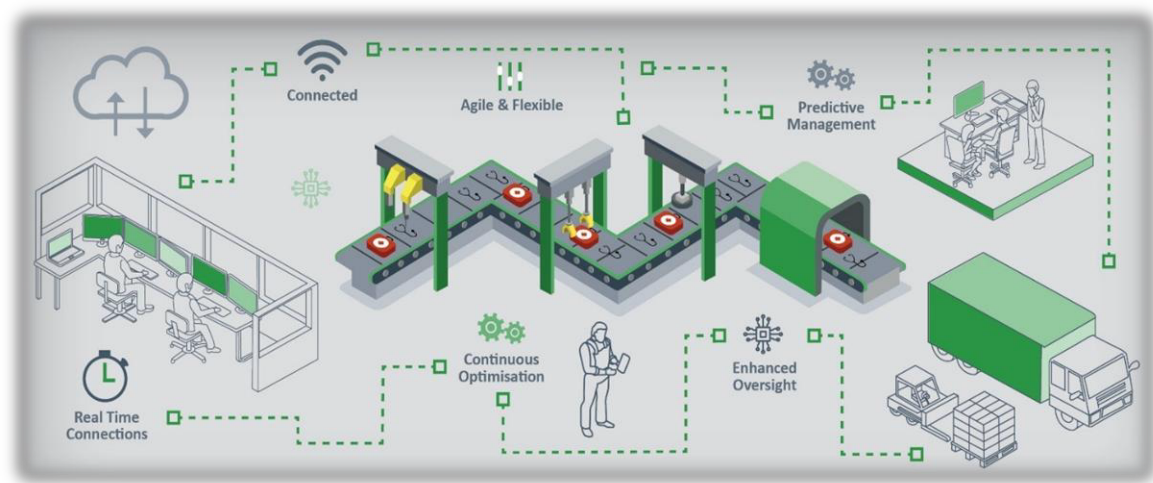


Figure 1 Industry 4.0 Solutions, available at:
<https://industry40thinktank.com/what-is-smart-factory-four-zero/>

2.5. ENVIRONMENTAL IMPACT AND SUSTAINABILITY

At the same time, there are urgent demands for the increased automation and connectivity of machines, and these also come along with pressing environmental impacts and sustainability concerns. On the one hand, effective machinery, intelligent energy monitoring, and optimized supply chains may just as severely reduce energy consumption while bringing material waste close to zero. The use of real-time accumulated data makes it possible for manufacturing processes to root out inefficiencies, minimize emissions, and contain over-exploitation of resources.

On the other hand, present-day industrial systems either are based on very complex infrastructures, including extremely large sensor networks, and also include the most sophisticated high-performance computing hardware-the combination of which produce today increasing total energy demand and electronic waste. Moreover, it has been a very

strong investment for factories in terms of an end-of-life management system for equipment upgrade and disposal. This is the aspect in which technology and ecology became priorities: renewable energy use, closed-cycle recycling systems, and ecological monitoring along the whole production line forever changed boundaries in technological development. Currently, productivity and sustainability abound much of the concern that much of today's various and forthcoming automated advances in industrial environments will share..

2.6. ONGOING TRENDS AND FUTURE OUTLOOK

Digitalization continues to promote and accelerate all innovations around artificial intelligence, machine learning and edge computing to make factories live and responsive with minimal human interference or intervention. Predictive maintenance algorithms for instance observe sensor data and predict mechanical faults to minimize:

Time-out-of-production and cost of operations; cloud-based remote monitoring; and digital twins with virtual prototypes reduce the cycle of development as well as the mandatory physical test. Further, production goals will include increasingly sustainability metrics, like the integration of renewable energy systems and efficiency improvements in resource use, to further minimize ecological damage. One of the primary foci is cybersecurity as the increasingly integrated machines and networks are the softest targets for malicious attacks. The working dynamic, it is safe to say, would drift towards higher-skilled jobs looking into an interpreting the systems data-driven. This is thus all going into the direction viewed in a high sense of smart automated factories which are not meant only to be efficient but also resilient-a veritable holy ground of technological evolution and responsible innovation.

3. THE WORKING ENVIRONMENT: CISCO PACKET TRACER

3.1. OVERVIEW OF CISCO PACKET TRACER

Cisco Packet Tracer is a packet simulation tool developed by Cisco Networking Academy. With this solution, students and even practicing professionals can play with topologies without having to have too much physical equipment at all. Progressing over time, the features of Packet Tracer improved from very high-level routing and switching to having some parts of IoT, which allows users to simulate sensors, actuators, and microcontrollers.

The overall GUI of this tool is very straightforward for the design of networks, configuration of devices, and visual representation of how data should flow from node to node. The measurable aesthetics make it pleasant to the total novice while deep enough for advanced users. There are several modes available, such as real-time mode (continuously running) and simulation mode (stepwise), where users can locate and debug bottlenecks from misconfigured devices to monitor performance under different network conditions.

3.2. KEY FEATURES AND CAPABILITIES

Probably the strongest attribute of Packet Tracer is its flexibility: it supports a wide range of network devices, from routers, switches, and firewalls to servers, extending to IoT devices such as temperature sensors, motion detectors, and programmable boards. Key functionalities include:

Interactive Network Building

Users can drag and drop network elements, such as routers, switches, and end devices, into the workspace and interconnect them with virtual cables. Each node's interface, IP addresses, protocols, and security settings can be configured.

Protocol Simulation

Basic routing protocols, such as RIP, EIGRP, OSPF, and BGP, are implemented. Packet Tracer also simulates some Switching-VLANs, Trunking, and Spanning tree-and security skills: it includes some topics like ACL, firewall, VPN concepts. That gives great possibilities of testing network configuration.

Integration with IoT

Beyond the ability to simply network them, Packet Tracer also has a number of IoT objects for which simple logic can be programmed: things like motion sensors, environmental monitors, and RFID readers. This is especially critical in those projects that involve smart environments and industrial automation.

Real-Time and Simulation Modes

- **Real-Time Mode:** It would be like working in a real-time environment, where the flow of traffic is continuous in the network and the configuration changes take effect immediately.
- **Simulation Mode:** The user can stop the movement of the packets and start going step by step for much easier paths of data analysis, easily debugging, and how devices respond upon some event that is to occur.

Together, all these features permit extensive testing: for even the most complicated designs—for instance, of an automated factory setup—it ensures checks way in advance of deployment in the real world. [3]

3.3. ADVANTAGES IN AN PROJECT CONTEXT

Some of the reasons why Cisco Packet Tracer is considered such a great teaching tool by everyone have been discussed below:

- **Price Effectiveness**

Networking and IoT training become largely convenient as institutions and learners can now practice network configurations instead of buying extensive physical hardware.

- **Safe Test Environment**

Packet Tracer provides a safe platform to try out protocols and make huge changes without damages or inoperative equipment or systems.

- **Instantaneous Feedback**

Changes now reflect almost at once, so students or project teams can iterate and refine their setups, gaining confidence in their configurations.

- **Collaboration and Learning**

With such easy sharing of Packet Tracer files, several users can work together on the same network or scenario, even from remote locations. This fosters teamwork as well as forms collective problem-solving groups for the students. Also, tutors can craft labs and challenges for guided exercises.

- **IoT Scenarios**

From simple-sensor-level basic sensors up to fully configured factory-floor layouts, it does incorporate catalogues of IoT devices required in industrial automation amidst the future Internet of Things.

4. METHODOLOGY

4.1. SYSTEM ARCHITECTURE

An intelligent and integrated framework of IoT-based sensors, actuators, and controllers across multiple functional safety, energy, material handling, and office facility zones proposes an automated factory. A common intelligent ecosystem is created by this zone interconnection taking place. Hosting the entire model in Cisco Packet Tracer gives it spatial qualities similar to a real-world environment as it follows the principles of Industry 4.0. Below is a summarization of his features and capabilities:

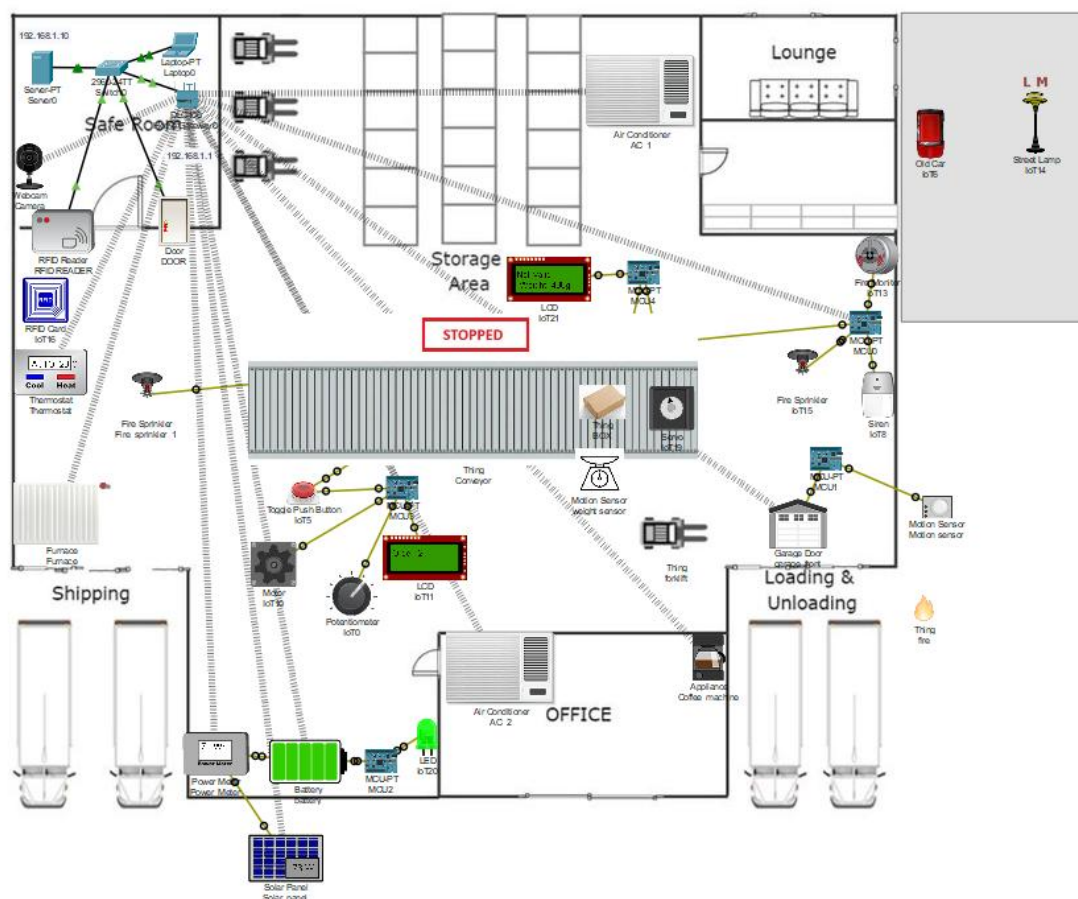


Figure 2 Factory Model in Cisco

The entire automated factory environment has been captured in a single span as showing in the Figure 2 using Cisco Packet Tracer. This design is intended to represent a realistic industrial environment with different critical zones such as a Safe Room, Storage Area, Office, Lounge, Shipping, Loading & Unloading bay; all of them having those relevant IoT devices, network components, and safety systems. Below is the brief outline of the major zones and some connected systems:

Safe Room

- Factory RFID readers and card scanners controlling door access.
- Includes IP cameras or webcams for security monitoring.
- Contains a dedicated laptop and server components for localized control and data logging.

Storage Area

- Thing Conveyor with IoT controllers for item automatic handling.
- LCD display IoT21 shows status messages such as Stopped or still active conveyor description.
- Sensors for motion will track active material movement to the record incoming and outgoing materials properly.

Loading and Unloading Bay

- Garage door management by a motion sensor (auto opens when cars are approaching).
- Other sensors such as fire sprinklers, sirens, etc., to manage safety rules even in high-traffic zones.

Office and Lounge

- Office area contains AC units (AC 1, AC 2) which are networked to IoT thermostat for monitoring of the temperature in real-time.
- The miscellaneous IoT devices which inhabit the lounge (coffee-machine, among others) bear witness to the automation for comfort.
- An old car object, aimed at illustrating the possibility of IoT sensors in vehicles- (for R&D use) is among those objects dedicated for IoT.

Energy Management

- Panel of solar energy along with a battery and a microcontroller that will keep a track on the power levels from consuming this to renewable energy.
- The storing of this renewable energy in addition to using the main grid less is what makes the above feature eco-friendly.
- A real-time charge status can be displayed by an LED.

Environmental & Safety Devices

- Fire sprinklers, alarm sirens caused by temperature or smoke sensors.
- Motion sensors installed in strategic locations (near doors or in the loading area) to automate gates.
- Climate control for storage areas and offices provided by thermostat devices.

Networking Backbone

- Most IoT devices communicate with a common network, have intercommunication with one another, and communicate with the control panel using switches or an IoT gateway.
- A dedicated server may handle advanced monitoring, data logging, and access control logic.

User Interfaces and Control Panels

- Operational status, temperature reading, and power levels are shown in different LCD displays.
- A main dashboard (usually a laptop or server interface in Packet Tracer) coordinates device logic, sends alerts, and manages automation scripts (e.g., adjusting AC units).

With this architecture, we can have an IoT ecosystem that is highly integrated, where sensors, actuators, and network controllers come to work in tandem. Every one of these functional areas-from safety to energy to the specifics of material handling or general office comforts-is tied into that singular monitoring and management environment. The sensors from that monitoring environment are pooled.

4.2. IMPLEMENTATION STEPS

Centralized control by an IoT Client (in this case a laptop) of all devices in the same network was implemented, and this is what the process looks like, along with some benefits of managing devices under a single interface.

4.2.1. *DEVICE DEPLOYMENT*

The first step involved putting all virtual devices such as MCU units, RFID readers, solar, and so on, at strategically designated zones such as Safe Room, Storage, and Office. An actual IP address was assigned to each device. Basic protocol configuration is done to facilitate communication without interruptions throughout the factory. Organizing the devices according to functional areas clarifies their identification and makes monitoring of each asset easier.

4.2.2. IOT SERVER

For example, a specific IoT Server was installed to centralize the collection and management of data concerning devices. This IoT server was accessed in real-time by IoT Monitor from Packet Tracer and provided the raw data like temperature or energy consumption readings and showed them into a single dashboard. Activation, deactivation, or configuration changes can also be done by the administrator on any other IoT device itself through that particular window, thereby eliminating the cost of maintenance.

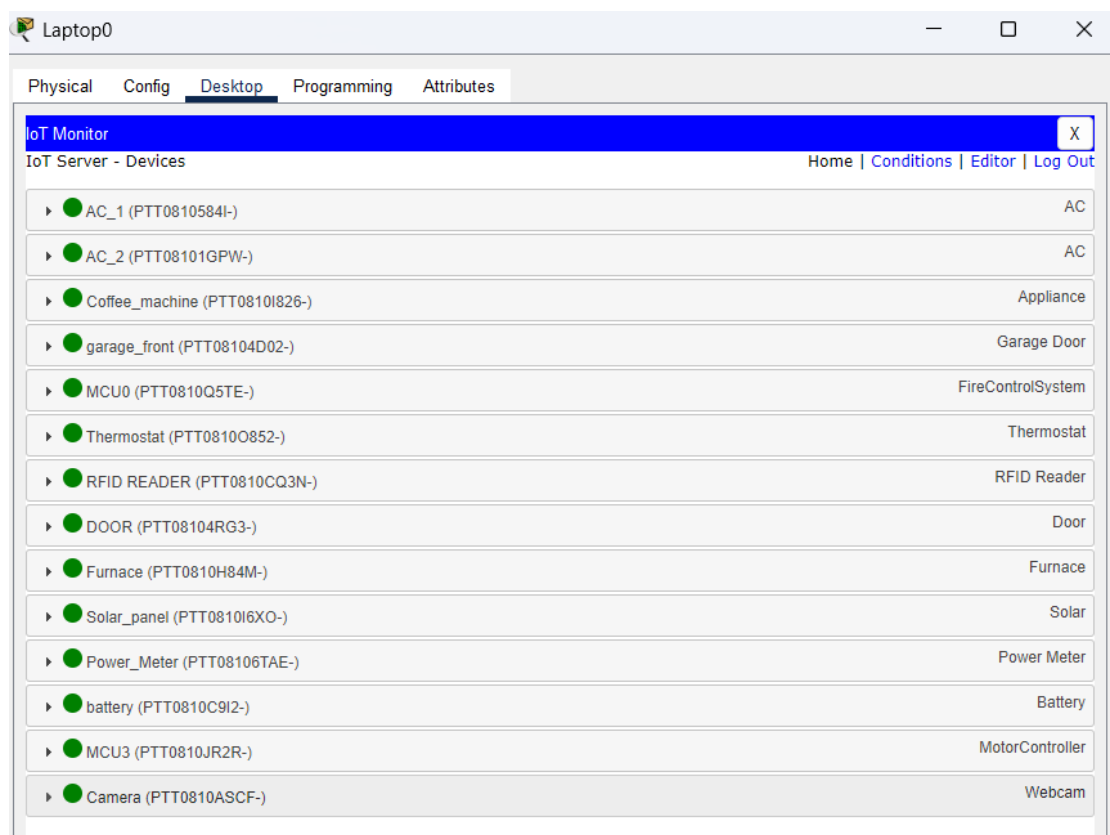


Figure 3 IoT client list of connected devices

4.2.3. *DEFINING RULES AND CONDITIONS*

One important rule in any automated ambient is that the rule and condition whether not IoT devices accept in some altering environment. In factories or centers, the IoT conditions can vary from climate controls activated with temperature triggers to access permissions managed by an RFID system. Thus, with a centralized editor in the IoT platform, administrators can swiftly set or modify any such condition in line with changing operational needs. In a practical context, rules might include:

- **Environmental Controls:** Automatically actions base on temperature (e.g., turn on air conditioning when temperatures rise above a specified threshold).
- **Security Measures:** Locking doors fails verification with certain RFID credentials or triggering cameras in an attempted breach of access.
- **Equipment Management:** Starting or stopping conveyors, furnaces or pumps, according to production schedules and sensor data.

Table 1 Conditions In Model

Name	Condition	Actions
Card_Valid	RFID READER Card ID = 1001	Set RFID READER Status to Valid
Card_Invalid	RFID READER Card ID != 1001	Set RFID READER Status to Invalid
Door_lock	RFID READER Status is Invalid	Set DOOR Lock to Lock
Door_Open	RFID READER Status is Valid	Set DOOR Lock to Unlock
AC_ON	Thermostat Temperature >= 25.0 °C	Set AC_1 On to true Set AC_2 On to true
AC_OFF	Thermostat Temperature <= 20.0 °C	Set AC_1 On to false Set AC_2 On to false
Furnace_ON	Thermostat Temperature < 16.0 °C	Set Furnace On to true
Furnace_OFF	Thermostat Temperature >= 25.0 °C	Set Furnace On to false
Camera_ON	DOOR Lock is Unlock	Set Camera On to true
Camera_OFF	DOOR Lock is LockDOOR Lock is Lock	Set Camera On to false

4.2.4. *WHY CENTRAL CONTROL MATTERS*

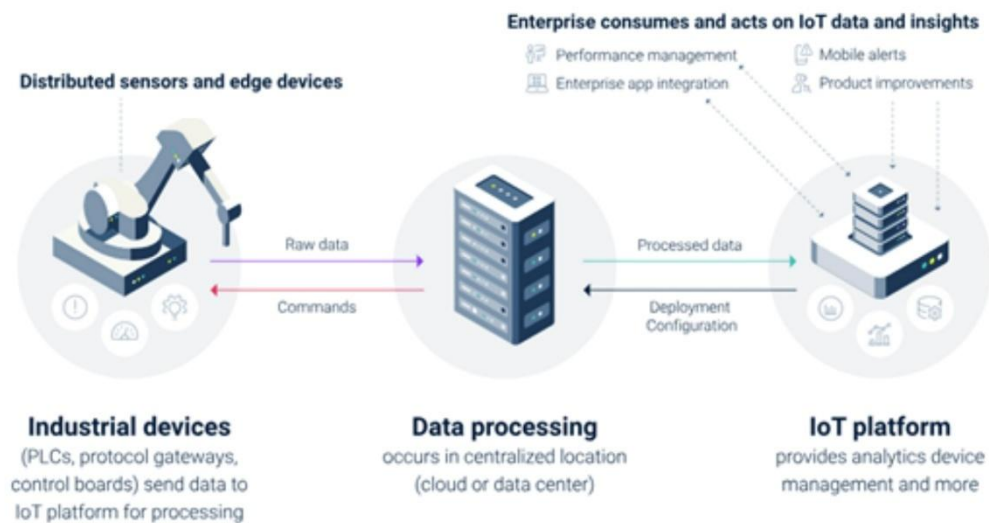


Figure 4 IoT Control model, available at:

<https://www.cumulocity.com/resource-library/the-internet-of-things-iot-a-practical-guide/>

Even data regarding door states or energy consumed at the respective points, everything can be seen within a framework for centralized control within a single interface. This would make diagnostics and decision-making a faster process. When an update needs to be made during operations, for example, changing temperature thresholds, adding new rules, and many more, it can be done in real time without disrupting any operations on a larger scale. Furthermore, additional IoT components can be registered within the system and will automatically be managed as soon as they are recognized by the central server. Hence, security policies will be consistent across the board as authentication and access rights are centralized. Unauthorized modifications to mission-critical devices are mitigated. This, therefore, creates central control for a more robust and flexible factory operation, where it can also run independently as each subsystem comes under the same, easily manageable framework. [4]

5. IMPLEMENTATION OF SYSTEMS

5.1. IMPLEMENTATION OVERVIEW

This chapter projects the concrete realization of the automated factory, which will demonstrate every system built for the purpose of making it fully integrated and manageable. One, the specific IoT devices, the configurations, and the logic registrations are established, according to the methodology explained in previous sections, to materialize the objectives of the project. This chapter will present environmental controls and security mechanisms, energy management and conveyor automation, and how each of these elements was configured, tested, and validated under the Cisco Packet Tracer ecosystem. This way, all systems will work in concert and reflect a coherent design following the modern principles of Industry 4.0.

5.2. CONVEYOR SYSTEM

5.2.1. *MOTOR SYSTEM*

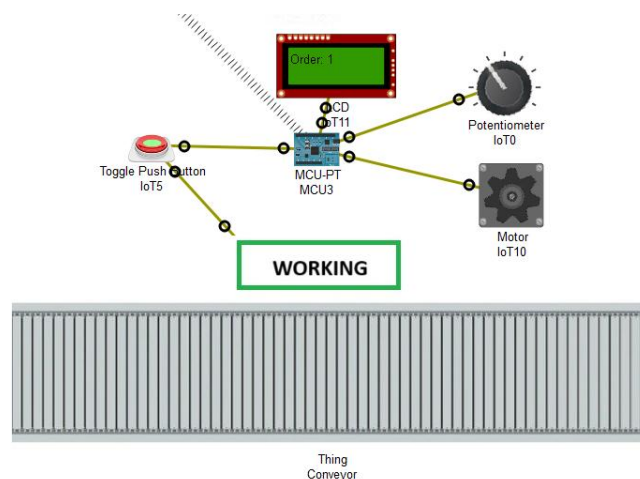


Figure 5 Conveyor system

The simple model for the conveyor system's control program makes it possible for microcontrollers to control conveyor performance by reading push buttons as start/stop trigger points and potential adjusting devices for speed and combined IoT relevance for remote monitoring. When started, the installation phase will initialize the relevant pins: LCD, button, potentiometer, and motor, as well as initialize the configuration of IoEClient object that will allow the microcontroller to be capable of reporting and getting state information from a central server. Adopting the update cycle of the main loop, the system checks at every cycle for the push button and, when pressed, increments its internal 'order' counter and displays this updated order value onto the LCD. Subsequently, it reads current potentiometer level so that the speed of the motor can be known and either energizes the conveyor at that speed or stops it altogether, depending on the pushes. Every update: with either new order value or changes in conveyor energy-speed, relayed to IoT server so operatives can get real-time data such as "THE CONVEYOR IS RUNNING" or its current speed through any remote interface. With such a two-way channel, continuous monitoring can be established while also allowing for remote overrides and conditional logic, linking the conveyor to industrial floors as part of more extensive processes. The system is also a fine example of how IoT enabled physical controls such as buttons and potentiometers that demonstrate simplicity in aligning hardware action tracking into production metrics and setting automation rules, which dynamically respond to sensor inputs or operational constraints across any industrial environment.



Figure 6 Information in Control System (Motor ON and OFF)

5.2.2. *WEIGHT SYSTEM*

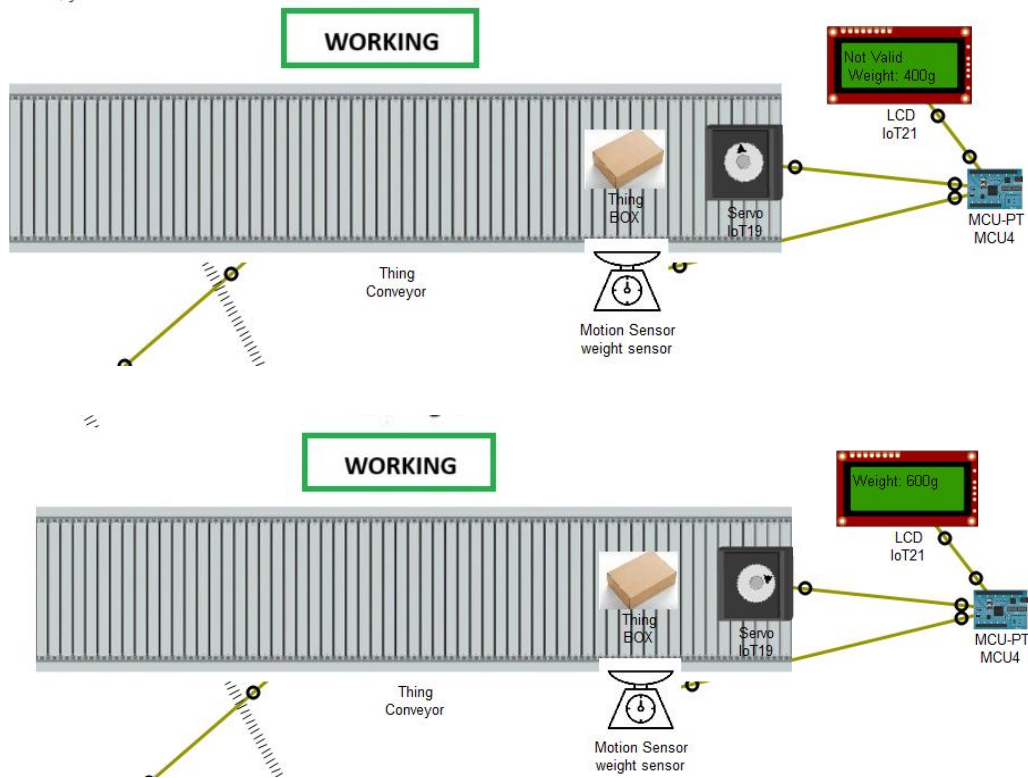


Figure 7 Weight Sensor

This is weight measurement system which senses the weight on the sensor and displays its measurement on an LCD while controlling a servo to show if the weight is within a usable range.. The code in MCU reads an analog value ranging from 0 to 1023 from the sensor, then translates it into a rough weight estimate (integer representation here), and checks if the weight falls within a valid threshold (550 to 700 grams). If valid, it updates the LCD with "Weight: Kg" and moves the servo to position 160 for packaging process; otherwise, it delivers a "Not Valid" prompt with the read weight and sets the servo position at 80 so the worker needs to check the reason of not valid weight. Such a logical, albeit simple, would enable real-time validation of the measured items so that only those within the required weight trigger the intended action of the display and servo output.

5.3. FIRE SYSTEM

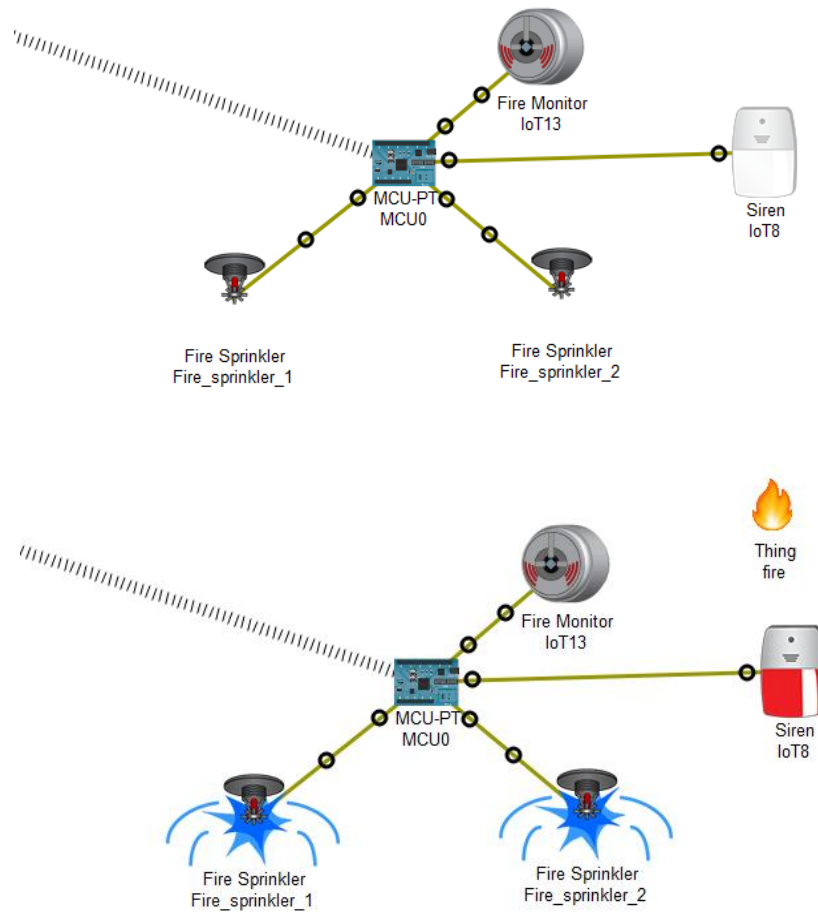


Figure 8 Fire System

This fire system script checks the Fire Monitor sensor and works with siren and sprinklers to respond to flames. When first initializing the program, the microcontroller uses signal from the fire sensor to determine whether any fire condition (in fact, to create a HIGH signal) exists. If true, it activates the siren and sprinklers to send ON signals and update internal variables (siren_status and sprinkler_status) for the same.

One of the important usage of this system is an integration with the IoT client. Microcontroller registers itself as FireControlSystem using the IoEClient to set up the siren and sprinkler statuses as states. On each iteration of the sensor check in the loop, these states report themselves back to the IoT platform; as such, IoT implements remote monitoring in real-time for any event involving fire. After the absence of flame, the siren and sprinklers

turn off, and the states report again so that the IoT interface reflects the site reality accordingly. Thus, this simple but effective mechanism allows instantaneous responses to fire hazards with oversight and logging capabilities as well as future remote overrides within a connected factory environment.

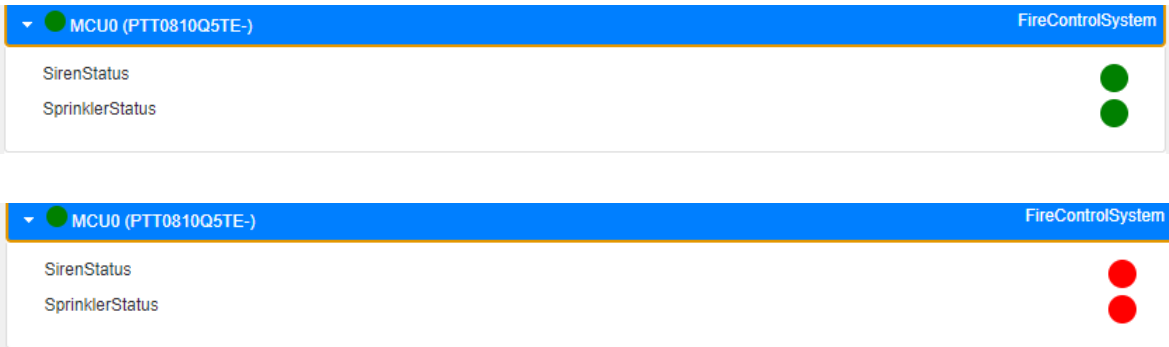


Figure 9 Information in Control System (Sprinklers ON and OFF)

5.4. SOLAR PANEL SYSTEM

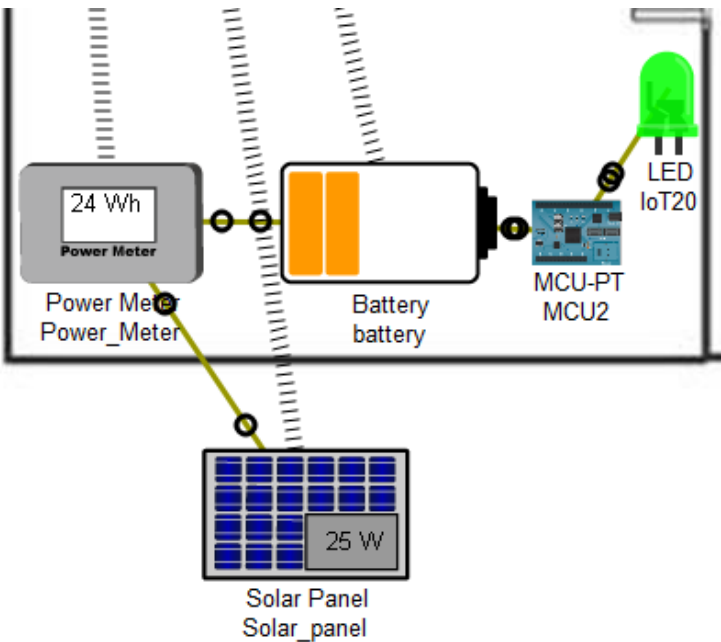


Figure 10 Model of Solar System

Using this subsystem, sunlight can provide reduced reliance on external electrical sources while creating an even better sustainable energy source for the facility. The electric energy

produced is stored in a power meter that measures the conversion of sunlight into electrical energy by the solar panel. This extra power would then charge the battery, ensuring backup storage levels even when the sun's rays may not be optimal or even more often consumed. Connected devices pass readings to an IoT platform for the managers to view in real-time what the current charge levels look like and how energy is utilized on a centralized dashboard. The energy that has been stored can be used to ensure essential services during a power outage or at times of excess demand. This would ensure an increase in the energy efficiency of the operation. It greatly improves the resilience of the systems since local solar production is integrated with cloud-based monitoring and battery storage and thus leads the factory to be ever more sensitive to the environment.

▼ ● Solar_panel (PTT0810i6XO-)	Solar
Status	36 Wh
▼ ● Power_Meter (PTT08106TAE-)	Power Meter
Status	34.719 Watts
▼ ● battery (PTT0810C9I2-)	Battery
Available power	56.5638 %

Figure 11 Status of Solar System IoT Monitor

5.5. TEMPERATURE CONTROL SYSTEM

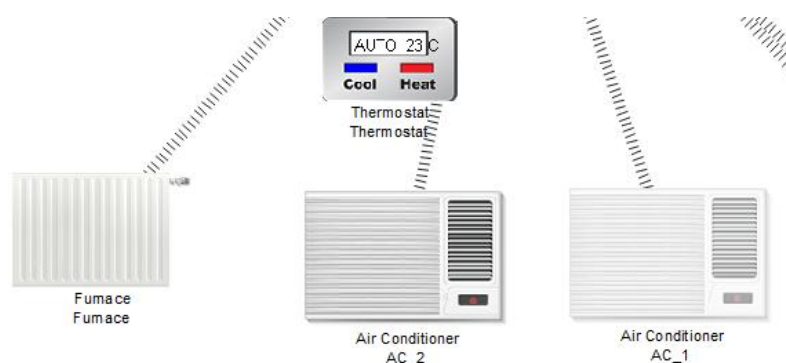


Figure 12 Model of Temperature Control System

This subsystem provides provisions for the observation of a furnace and air conditioners managing the internal temperature. Thus, within a centralised thermostat, ambient

temperature is measured continuously and is then found to decide whether to cause cooling and turning on of the A.C units or heating by a furnace. Whenever the temperature exceeds a limit, the air conditioners automatically switch on; in contrast, when temperature is recorded below a lowering limit, the furnace heating comes on. Presently, a microcontroller transmits real-time status and actual temperature into the IoT platform also with heating-cooling mode and energy consumed contexts for the user to remotely observe and even fine-tune the thresholds for temperature set. Temperature Control System dynamically adapts to the surrounding environmental conditions to provide optimum climate control without associated wastefulness in energy consumption.

5.6. RFID CARD SCANNER

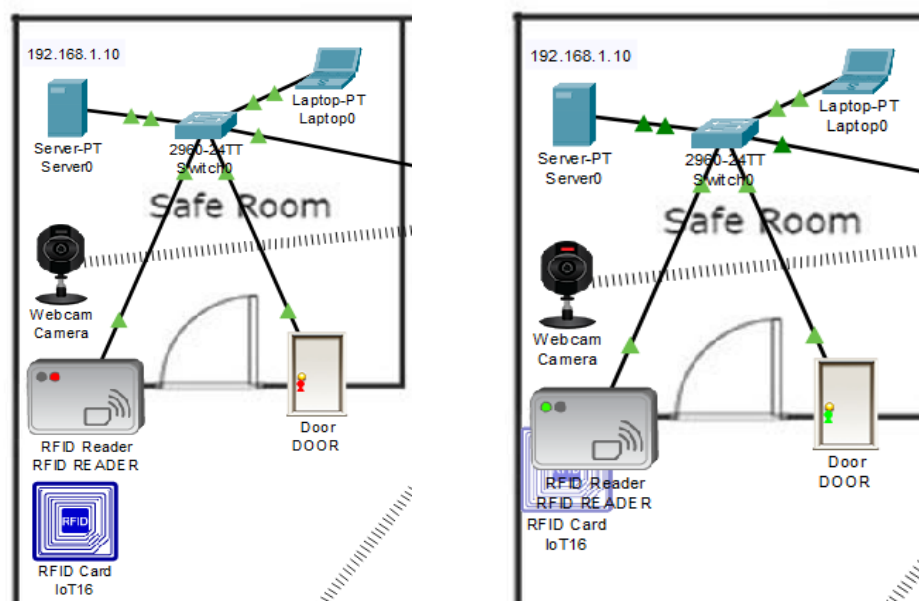


Figure 13 RFID Card Scanner

With the RFID card scanner system, the entry to the server room is to be limited as provided in the previous condition that it unlocks the door only when a valid RFID READER Card ID is presented, such as 1001 in the example. Once recognized by the scanner as one of its trusted credentials, it signals the door to unlock and, most probably, activates a webcam for

the purpose of security logging. An invalid or unknown card causes the system to lock immediately and turn off the camera. This is done in order to keep resource optimization and privacy considerations in mind at the same time. The system precisely ensures that only authorized individuals are physically permitted access to critical infrastructure, thus reducing the probability of tampering or data breaches in the wake of unauthorized access. By coupling the RFID reader output to IoT events, administrators can also access and track door activity remotely, controlling who accesses the server room, as well as updating permissible card ID access immediately after the induction of new personnel or changes in security policy.

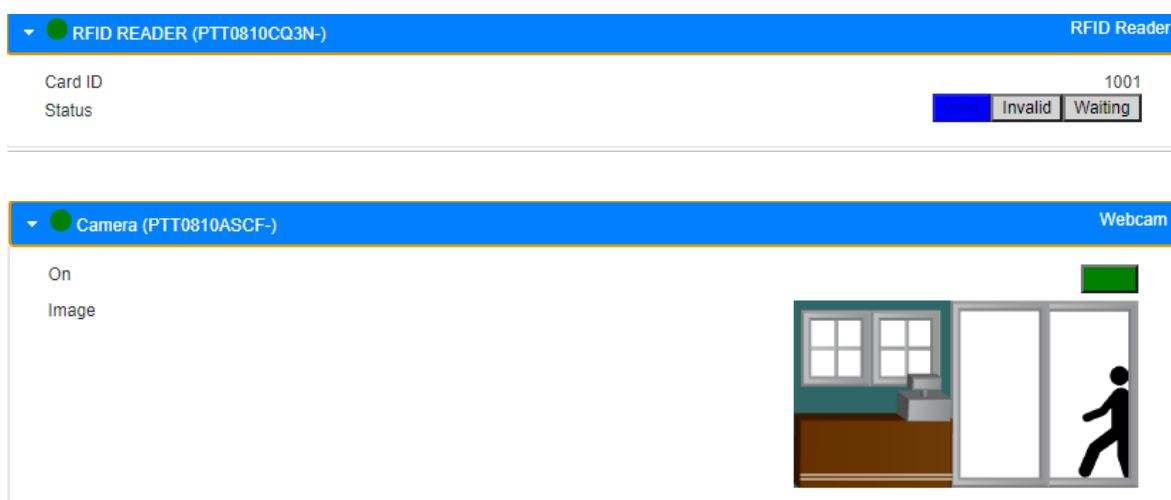


Figure 14 Status of RFID Scanner in IoT Monitor

5.7. AUTOMATIC GARAGE DOOR

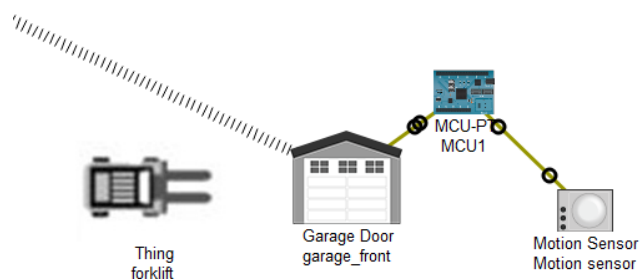


Figure 15 Garage Door Model in Cisco

This household subsystem might be annoying or irritating because it uses a motion detector and microcontroller to open a garage door when it detects movement nearby and closes it

the moment no movement is detected. For example, in this case, a forklift goes around periodically updating an internal parameter thinking it is moving and thus causing the motion sensor to pick something only when it is inside range. After the "movement detected" signal from the sensor has been received, the microcontroller send signal to open the garage door to allow entry or exit of the forklift without manual intervention. After movement is no longer detected by the sensor, the microcontroller could either trigger automatic closing of the door or maybe wait for a set timeout period to be eco-friendly both in terms of energy and manual effort. With the above demonstration, it becomes clear that such a system from simulated movement of a forklift to input by the sensor can, through real-time detection, automate actions that could otherwise consume functional time in manual interaction, thus streamlining logistics and somewhat boosting efficiency overall in the industrial environment.

6. NETWORK AND SECURITY

6.1. INTRODUCTION

Robust network security becomes a vital prerequisite in industrial connected environments nowadays. This project makes provisions for multiple-tier defense measures in the system to safeguard data, devices, and operations from possible attacks. From the secured wireless covered by WPA2-PSK to network segmentation provided by the VLANs and ACLs, each mechanism contains an invasion and reduces unauthorized access. Furthermore, encryption and secure protocols minimize interception while centralized logging and possibly intrusion detection measures allow real-time monitoring of suspicious activity.

These features are representative of industry best practices as found in real-life situations in Industry 4.0, where IoT devices have to be highly productive and secure effects from diverse machines. It is in this chapter that while role-based access, secure key provisioning, and continuous system monitoring are presented, the importance of these measures for resilience between critical smart factory components, such as sensors, actuators, PLCs, and servers, will be eluded upon. While implementation currently emphasizes the aforementioned principles, these can be extended to incorporate more elaborate cybersecurity frameworks in future, hence ensuring that whatever the nature of transformation in manufacturing processes, the network and security architecture will continue to be robust and adaptive.

6.2. COMPARING THE MAIN WI-FI SECURITY STANDARDS

Table 2 Network Standards

Standard	Release Year	Encryption	Session Key	Authentication	Level of Security	Weakness
WEP	1997	RC4	64/128 bit	Open system, shared key	Very Low	Insecure encryption easily exploited by attackers
WPA	2003	TKIP / RC4	128 bit	Pre-shared key (PSK)	Low	Weak encryption, potential compatibility issues
WPA2	2004	AES-CCMP	128 bit	Pre-shared key (PSK)	Moderate	Vulnerable to KRACK (key reinstallation) attack
WPA3	2018	AES-CCMP / AES-GCMP	128/256 bit	Simultaneous Authentication of Equals (SAE) / PSK (Enterprise variants)	High	Complex deployment; higher resource requirements

WPA2-PSK, or WPA2-Personal, remains the most relevant standard with respect to varied environments and deployments, including this project. It balances all three criteria: security, compatibility, and ease of configuration - in case of mid-sized or small industrial networks; WPA3 introduces better encryption and authentication capabilities but would easily cause hitches during adoption because of its complicated setup and limited backward compatibility. However, WPA2-PSK:

- Still securing AES-CCMP encryption, which is deemed strong for most applications,
- Minimal to entirely none hardware-software extra updates demand, thus making it easier to roll out on a whole network of IoTs.
- Long and well-established track record with known best practices on how to mitigate such vulnerabilities as KRACK, for example, regular firmware updates and proper management of keys, show that it is documentation. [5]

Thus, this secure wireless environment is achieved without the tangles and hardware stress of moving to the latest standard, thus making it the best solution for many Industry 4.0 labs or small-to-medium factories possible.

6.3. ENCRYPTION AND SECURE PROTOCOLS

Adds In a setting where IoT gadgets, networks, and manufacturing systems talk a lot with one another, the importance of encryption and secure protocols could not be stressed enough in assuring the integrity of the information and preventing unauthorized access to it. This project takes a multi-level approach in ensuring that data is protected both during the transmission of that data and at rest. Following are the key encryption techniques and secure protocols that will improve the security of the network:

6.3.1. DATA ENCRYPTION

- **AES-Based Encryption**

WPA2-PSK is implemented at the wireless layer; thus, the communication is being protected using AES-CCMP encryption. AES (Advanced Encryption Standard) now remains an industry-standard security method providing robust security against most brute-force attacks. Thus, this guarantees that any information traveling over the wireless network, for example, a reading machine or operational commands, is less susceptible to interception or deciphering by malicious actors. [6]

- **Optional End-to-End Encryption**

Though WPA2-PSK would encrypt traffic at the link layer, few IoT devices or server communications may apply any other application layer encryption for critical information transmissions (e.g., TLS/SSL). Adding other encryption methods fortifies confidentiality so that even in the event of invasion into one of the layers, the information would either remain garbled/truncated or inaccessible at the other layer.

- **Key Management and Rotation**

To counter the threats posed by key reinstallation attacks (like KRACK), administrators keep changing pre-shared keys from time to time. This is a really simple measure to stop hackers from abusing the same key at extended intervals, thus making any compromised credentials become invalid within a short time.

6.3.2. SECURE PROTOCOLS

- **HTTPS for Web Interfaces**

As applicable in any web-based control panels or dashboards, HTTPS encrypts user credentials and other data exchanged from a client (for example, laptop or mobile device) to the server. If enforced through HTTPS connections, critical user information such as login passwords, commands to devices, or even performance metrics shall not be intercepted easily.

- **SSH for Remote Administration**

This is when administrators need to get access for terminal function into the network devices and servers; and the Secure Shell (SSH) technology is put into use. SSH sessions provide the configuration, firmware, and troubleshooting functions via a secured channel compared to previous plaintext protocols such as Telnet.

- **Device Authentication**

IoT endpoints are sometimes assigned specific credentials or certificates to improve authentication capabilities. In addition to focusing on WPA2-PSK and shared secrets, its advanced configurations can also include Public Key Infrastructure (PKI), thereby allowing each device the authentication of its identity toward the network through digital certificates.

7. RESULTS

Throughout this project, a number of results have emerged concerning the overall feasibility and applicability of the proposed solutions. Particularly important is the integration of an IoT device-from conveyor and weight sensors to fire systems and solar panels-which brought under management through Cisco Packet Tracer and shown to function well under simulated industrial conditions.

A well-coordinated activation performed by the conveyor subsystem reflects the direct response to user interaction by incrementing order numbers while varying the speed using a simple push button coupled with a potentiometer while providing accurate, real-time readings crucial for triggering within the limits imposed on the servo mechanisms for weight sensor. Last but certainly not least, the fire safety measures (Siren and sprinklers) have been duly confirmed with reliability that makes for the securing of the entire factory upon sensing any impact signal. The solar panel subsystem would thus prove potential for energy optimization by keeping track of power generated and battery storage. In addition, temperature control would be managed and would operate either the furnace or the AC depending on threshold levels without human oversight, thus lessening operator involvement. Lastly, critical areas were preserved for authorized personnel only by the use of an RFID Card Scanner and automatic garage doors, thereby minimizing unnecessary idle or unsafe movements inside the facility.

These measures were, strictly speaking, functional measures; in each area where the project team had realistic goals defined, they could be expected to achieve them. Communication between devices, linking to IoT servers, synchronization of triggers across subsystems, and work flow continuity therefore all proceeded with minimal delays or cuts.

8. CONCLUSIONS

As this project draws to a close, the main conclusions and their impact on the development options adopted need to be restated. The simulation environment, however, was a realistic interrelationship of industrial processes, energy management, and security mechanisms, hence, the cohesive value of IoT-based architecture in Industry 4.0 settings can be demonstrated.

Like always, certain limitations naturally influenced the throughline of the work. For example, some aspects of the learning curve associated with the many IoT components configured in Cisco Packet Tracer delayed deeper probing into advanced modalities of correlation or predictive analytics as the system could handle triggers and data flows. However, even at this stage, there seems to be wide scope for complex alarm correlation, root-cause analysis, or scalability tests under larger loads.

However, converting events from devices (sensor signals, for instance) into active tasks has vastly improved development productivity. Quite simply, similar to surprisingly fast processing of network alarms in real-world contexts, this infrastructure facilitated the rapid melding of new devices to rules and relatively simple validation against user expectations. Equally relevant, each subsystem (for example conveyor, temperature control, fire safety) was validated against the operational parameters envisioned for semi-automated factory scenario.

Well, from performance perspective, the tests conducted were exactly according to the expectations. As mentioned earlier, the systems managed a constant flow of events, whether be for logging and triggering security alerts or for updating energy usage, without showing significant latency during normal operational conditions. It is worth mentioning that actual field deployment will predetermine unique complexities in this scenario, but the work has demonstrated quite a number of practical applications in further work plans.

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